Experiment 7

Aim

To apply **K-Means and Hierarchical Clustering algorithms** on a customer dataset to group similar customers based on their characteristics.

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1. Dataset Source

Source:
https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python

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2. Dataset Description

The dataset contains customer information collected from a mall.

- **Total Records**: ~200 customers
- **Features**:
  - Customer ID
  - Gender
  - Age
  - Annual Income (k$)
  - Spending Score (1–100)
- **Target Variable**:
  No target variable (Unsupervised Learning)
- **Description**:
  The dataset is used to perform customer segmentation based on income and spending habits. The spending score is assigned based on customer behavior and purchasing data.

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3. Mathematical Formulation of the Algorithm

K-Means Clustering

Objective: Minimize intra-cluster variance

$$J = \sum_{i=1}^{k} \sum_{x \in C_i} ||x - \mu_i||^2$$

Where:

- C_i = cluster i
- μ_i = centroid of cluster i
- x = data point

Hierarchical Clustering (Agglomerative)

Distance between clusters:

- Single Linkage:

$$d(A, B) = \min(d(a, b))$$

- Complete Linkage:

$$d(A, B) = \max(d(a, b))$$

- Average Linkage:

$$d(A, B) = \frac{1}{|A||B|} \sum d(a, b)$$

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4. Algorithm Limitations

K-Means Limitations

- Requires predefined number of clusters (k)
- Sensitive to outliers
- Works best with spherical clusters
- Not suitable for non-linear data

Hierarchical Clustering Limitations

- High computational cost for large datasets
- Difficult to choose optimal cut level
- Sensitive to noise and outliers

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5. Methodology / Workflow

1. Import dataset
2. Select relevant features (Income & Spending Score)
3. Data preprocessing (scaling if needed)
4. Apply K-Means clustering
5. Use Elbow Method to find optimal k
6. Apply Hierarchical Clustering
7. Plot dendrogram
8. Visualize clusters
9. Compare results

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6. Performance Analysis

Since clustering is unsupervised:

- **Elbow Method** → Finds optimal clusters
- **Silhouette Score** → Measures cluster quality

Interpretation:

- Higher silhouette score = better clustering
- Clear cluster separation indicates good segmentation

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7. Hyperparameter Tuning

K-Means

- Number of clusters (k)
- Initialization method (k-means++)
- Max iterations

Hierarchical

- Linkage method (single, complete, average)
- Distance metric (Euclidean, Manhattan)

Impact:

- Correct k improves accuracy
- Proper linkage improves cluster quality

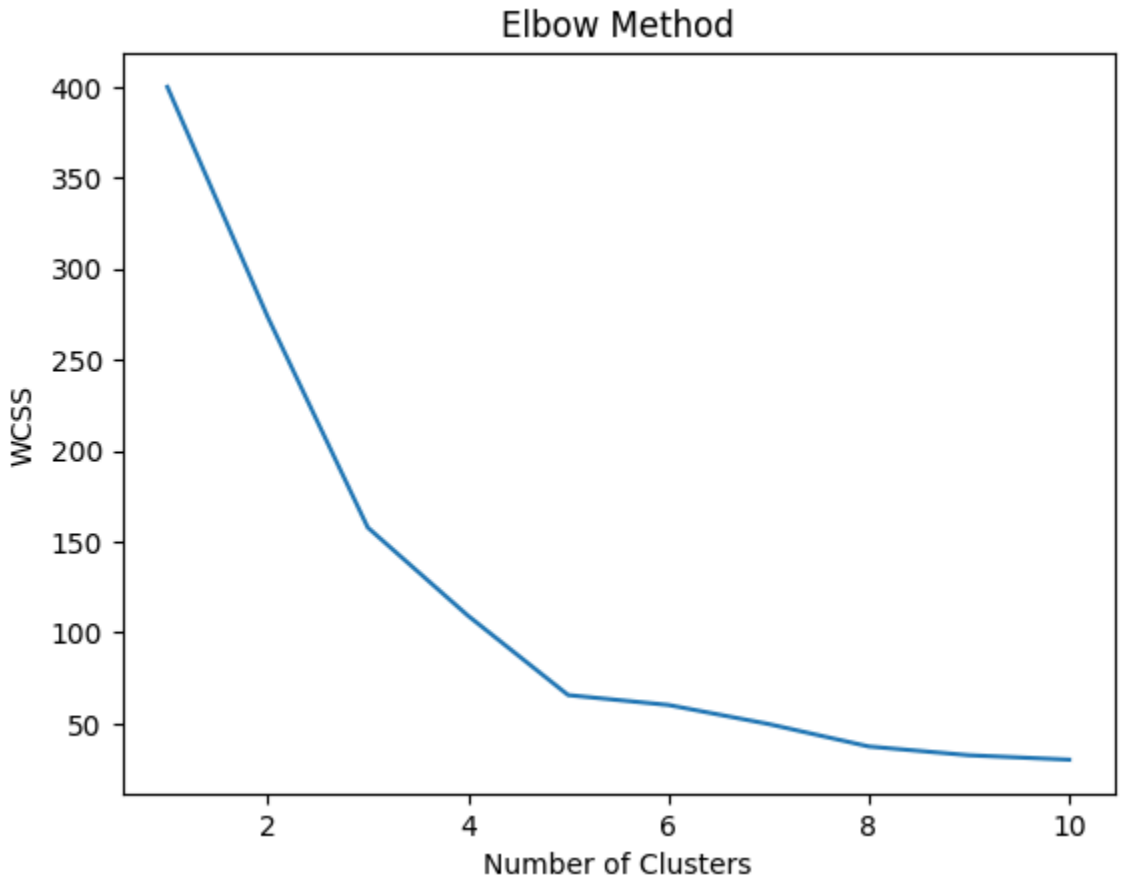
Code

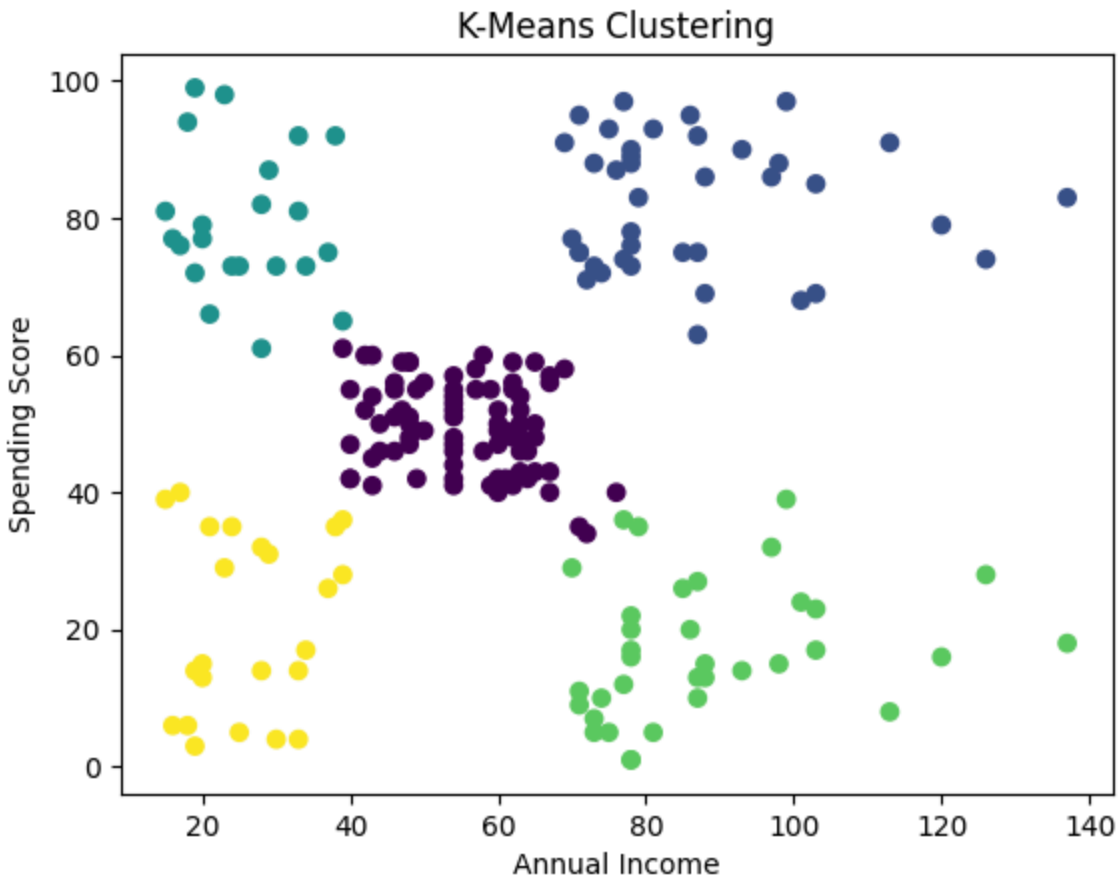
```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.cluster import AgglomerativeClustering
from scipy.cluster.hierarchy import dendrogram, linkage

# Load dataset
data = pd.read_csv("Mall_Customers.csv")

# Selecting features
X = data.iloc[:, [3, 4]].values
```

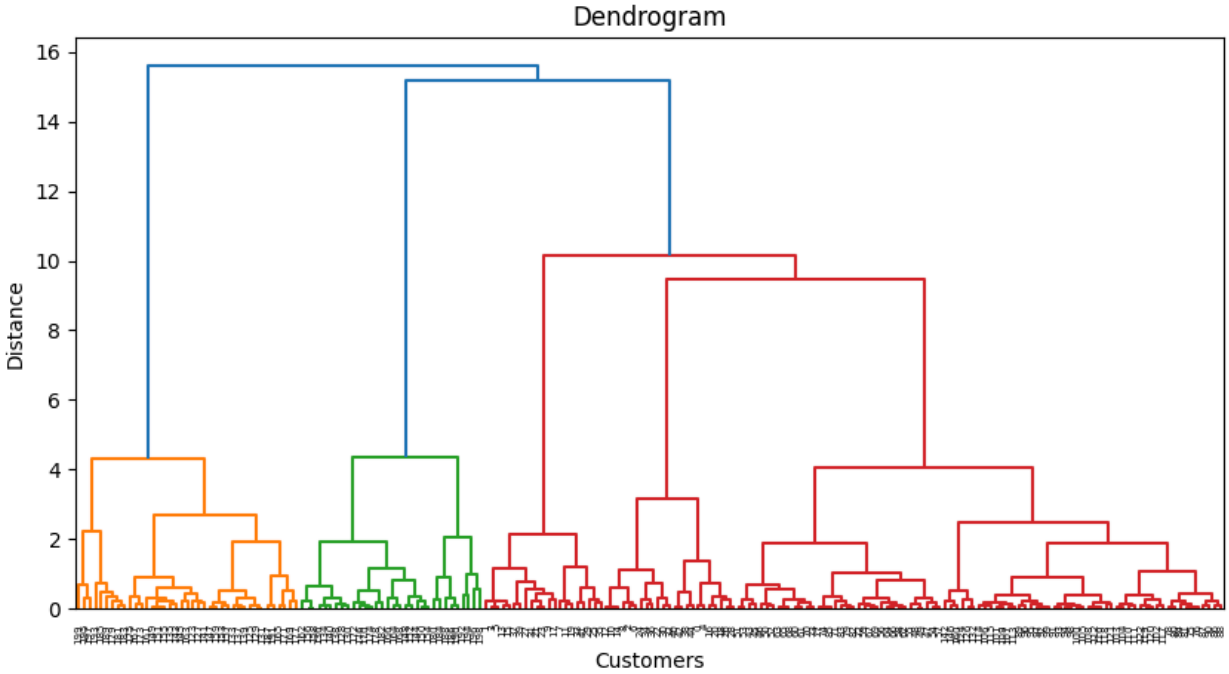
```
# ──────── K-Means ────────
wcss = []
for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=0)
    kmeans.fit(X)
    wcss.append(kmeans.inertia_)

# Plot Elbow
plt.plot(range(1,11), wcss)
plt.title("Elbow Method")
plt.xlabel("Clusters")
plt.ylabel("WCSS")
plt.show()

# Apply KMeans
kmeans = KMeans(n_clusters=5, random_state=0)
y_kmeans = kmeans.fit_predict(X)

# Plot clusters
plt.scatter(X[:,0], X[:,1], c=y_kmeans)
plt.title("K-Means Clustering")
plt.show()

# ──────── Hierarchical ────────
Z = linkage(X, method='ward')

# Dendrogram
dendrogram(Z)
plt.title("Dendrogram")
plt.show()

# Apply Hierarchical Clustering
hc = AgglomerativeClustering(n_clusters=5)
y_hc = hc.fit_predict(X)
```

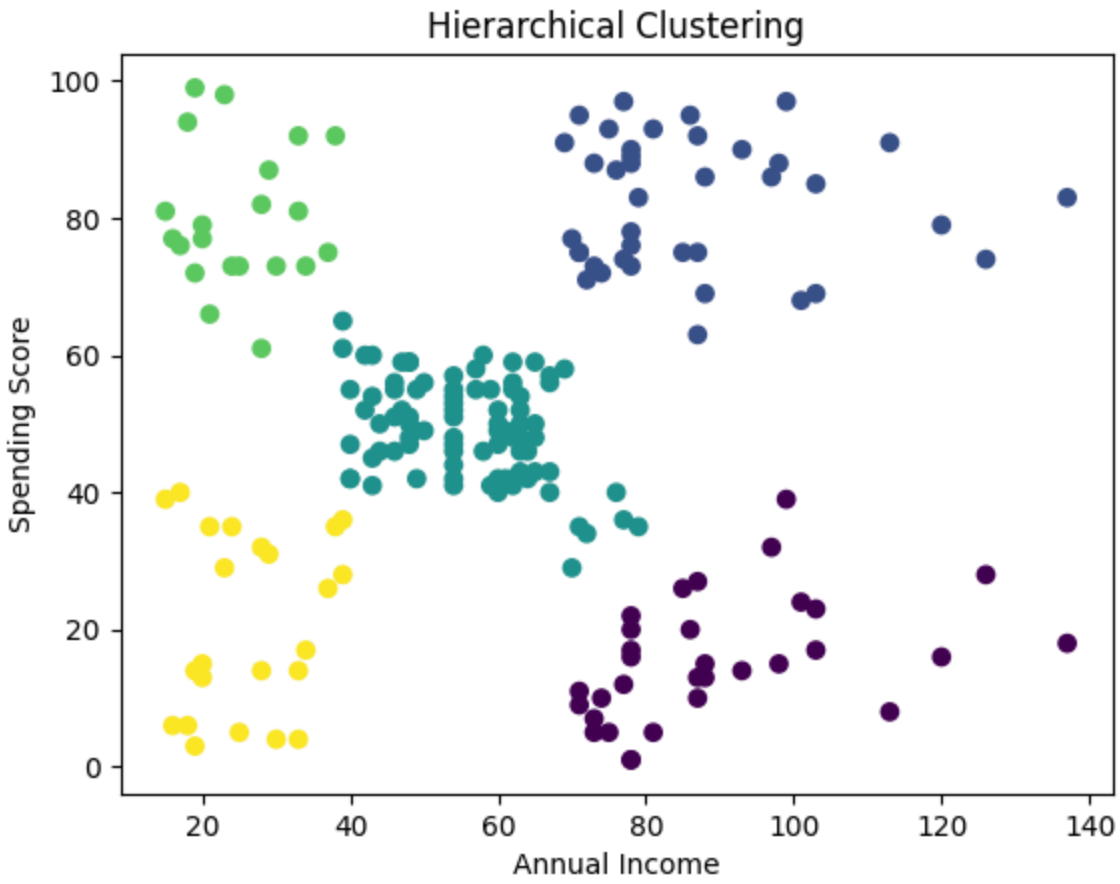
```
# Plot clusters
plt.scatter(X[:,0], X[:,1], c=y_hc)
plt.title("Hierarchical Clustering")
plt.show()
```

Output




K-Means Clustering
Spending Score
Annual Income
0
20
40
60
80
100
20
40
60
80
100
120
140


Dendrogram
Distance
Customers
0
2
4
6
8
10
12
14
16



Conclusion

K-Means and Hierarchical Clustering were successfully applied to segment customers.

K-Means provided faster results, while Hierarchical Clustering gave better visualization through dendrograms.

These techniques help businesses identify target customers and improve marketing strategies.